Name ______________________________ Section ________

MTH:2310 DISCRETE MATH
QUIZ 3
DUE: THURSDAY SEPT. 25

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: *(4 points)*
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 2 points for arithmetic errors.
You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. *(10 points)*
Use a direct proof to prove the following claim.

"For any integers x, y, and z, if x^2 divides y and y^3 divides z, then x^6 divides z.
Def: Given two integers b and a, if $b \neq 0$ we say b **divides** a if there is an integer c where $a = cb$ (or equivalently $\frac{a}{b} = c$).

Problem 2. *(10 points)*
Use a proof by contraposition to prove the following claim.

"If x and y are integers and $x^2(y^2 - 2y)$ is odd, then x and y are odd."

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Problem 3. *(8 points)*
Use a direct proof, proof by contraposition, or proof by contradiction to prove the following claim.

"If n is any integer and n^2 is divisible by 4, then n is odd."

Problem 4. *(10 points)*
Use a proof by contradiction to prove the following claim.
"Given any integers x, y, and z, if $x^2 + y^2 = z^2$ then x or y is even."